

**ADMINISTRATIVE INFORMATION****FMI No: 25392****FMI Title:** ORP2 and ORPV LSCOM FPGA Code Update

FMI TYPE	CHARGE CODES	IMPLEMENTATION DATES
☒ Mandatory Safety	GEMS - AM:	Issue Date:
☐ Mandatory	GEMS - E:	
☐ Mandatory on Request	GEMS - A:	Close Date:

☒ Staged FMI	Customer List provided by FMI Administration. If you identify additional units within effectivity range, but they are not shown on the customer list, contact FMI Administration in your Pole. ☒ Parts and any special tools are furnished automatically. To address shortages and ordering exceptions, contact FMI Administration in your Pole. ☐ Parts Ordering: GEMS - Americas, Contact your Region logistics coordinator GEMS - Europe, Order FMI kits from GPO/EPL Buc GEMS - Asia, Contact your region logistics coordinator ☐ No Parts are required to complete this FMI.
☐ Batch with	To minimize costs, combine implementation of this FMI with the FMIs listed for all customers where the effectivity is the same (Please see your customer lists). To minimize costs, combine implementation of this FMI with the FMIs listed for all customers where the effectivity is the same (Please see your customer lists).
☐ Non-Staged	No Customer List is provided. Each service location must identify affected units, and place part orders as indicated in the Special Instructions.

Technical Support

GEMS - Americas	GEMS - Europe	GEMS - Asia
GE Medical Systems Insite Center 800-321-7937 Milwaukee, WI 53201, USA	GEMS - E / ESC BP34 F 78533 BUC CEDEX, FRANCE Tel: (33 1) 39 20 00 07 or ESC Fax: (33 1) 30 70 99 70	Asia Support Center Phone: 81-426-56-0033 Fax: 81-426-48-2905

Special Instructions:

To save time at the site: Loading the Xilinx ISE Programmable Logic Device Software (included on CD-ROM disk with FMI kit) can be done prior to arriving on site. This software is required to be loaded on laptop and is used to flash program the ORP board. The software loading process needs performed only one time and can remain on laptop to be used at multiple sites as required. Once all sites that have been dispatched have had FMI 25392 loaded, one may elect to deinstall this software. See Appendix I for load instructions.

Leave the Xilinx ISE Programmable Logic Device Software and Programing Cable (FMI Kit) with system when finished. Secure with other system tools.

Unit Tracked: Refer to Effectivity list on page 2.**FMI Instructions Part No.:** 5193722-100. Rev 3**Do NOT leave unsecured on site**

FMI Completion Certification Report

Mail To

GEMS - Americas	GEMS - Europe	
G.E. Medical Systems Attn: FMI Admin. W-523 P.O. Box 414 Milwaukee, WI 53201	GEMS - Europe FMI Admin. 322 BP34 F78533 Buc Cedex, France	Send to your Country FMI or Service Administrator

FMI No.: 25392

FMI Title: ORP2 and ORPV LSCOM FPGA Code Update

Customer Full Name: _____

Number and Street Address: _____

City and Country: _____

Country Code

Site Number

System I.D.

Unit Tracked:

Model Number	Serial Number	Compl. Date DD/MM/YY	FMI Comp. Code	Service Hours	
				Labor	Travel

FMI Completion Code

1. Modification Complete or previously modified.
2. FMI not required per FMI instruction.
3. Not modified. Unit in storage. FMI kit returned to stock.
4. Modification refused or equipment altered. Customer signature and title required.
5. Unit location unknown or unit not in area. Product Locator notified

Customer Refusal or Equipment Altered - Modification Pending.

Installer

Comments

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Revision History

Revision	Date	Reason for Change
1	November, 2006	Initial release
2	December 12, 2006	Updated for review feedback for milestone "FE"
3	February 26, 2007	Updated for review feedback from Pilot, Milestone "F3"

Effectivity

All CT systems that utilize the H-power or VCT gantries with rotation speed of 0.4s or quicker.
Model Numbers:

VCT Systems

Gantry Models for ORPV Boards	Description
5124069	VCT 64
5126093	LightSpeed Pro32

LightSpeed Pro16 Systems

Gantry Models for ORP2 Boards	Description
2357739	LightSpeed Pro16 (HP-100)
2357739-3	LightSpeed Pro16 (HP-80)
2357739-4	LightSpeed Pro16 (HP-100)
2357739-6	LightSpeed Pro16 (HP-100).
2357739-7	LightSpeed Pro16 (HP-100)
2357739-8	LightSpeed Pro16 (HP-100)

Note: Do not assume the gantry contains a specific version of ORP board. It is important to physically check and identify which ORP board resides in system. While most HP80/100 systems were manufactured with ORP1 boards, over time they have likely been upgraded to ORP2 boards.

Please go to Section I, Gantry Preparation and ORP Identification (page 6) and follow the steps to identify which ORP Board version is on your system. Table with all versions listed in on page 8.

Purpose

The purpose of this FMI is to address the following circumstances.

Numerous PSR and PQR reports have been entered in iTrak for systems seeing "kV loss", "-1999 mA value" and missing scan data.

Due to frame timing code error in the Xilinx - Field Programmable Gate Array (LSCOM FPGA) of the ORP1-4, ORP2-2 and ORPV boards; a slip ring disconnect signal occurring along with a DAS trigger may result in an ORP command to turn off the High Voltage at the Jedi generator. The occurrence of this ORP frame timing error is proportional to the amount of slip ring disconnect signals and the rotational speed (trigger rate) of the gantry. Consequently the above-mentioned faults happen most often on VCT/PRO32 systems rotating at 0.35/0.40s, followed by HP80 & 100 systems rotating at 0.40s.

Since no iTrak reports have been submitted for the above-mentioned faults on systems rotating at 0.5s or slower, these systems have not been included in the FMI effectivity. Field FRU stock will be updated for support of the installed base.

Prerequisite FMIs

None

Related FMIs

None

Furnished Materials

FMI 25392 Kit part number: 5193722

	Qty	Part Description	Part Number
1	1	FMI Software CD-Rom	5196105
2	1	ORP Board Labels	5196108
3	1	Xilinx Programmer I/F Cable	5196124
4	1	Xilinx Platform Cable (USB)	5196130
5	1	FMI Instructions	5193722-100

Tool Requirements

Quantity	Description
1	FE Toolkit
1	ESD wrist-band
1	2.5mm Hex Key
1	5mm Hex Key
1	Standard Straight Blade Screwdriver

Personnel Requirements

Resource	1 FE
Labor	1.5 hrs
Travel	1 hr

Before You Begin

Before beginning this procedure, please read the gantry safety information.

Safety



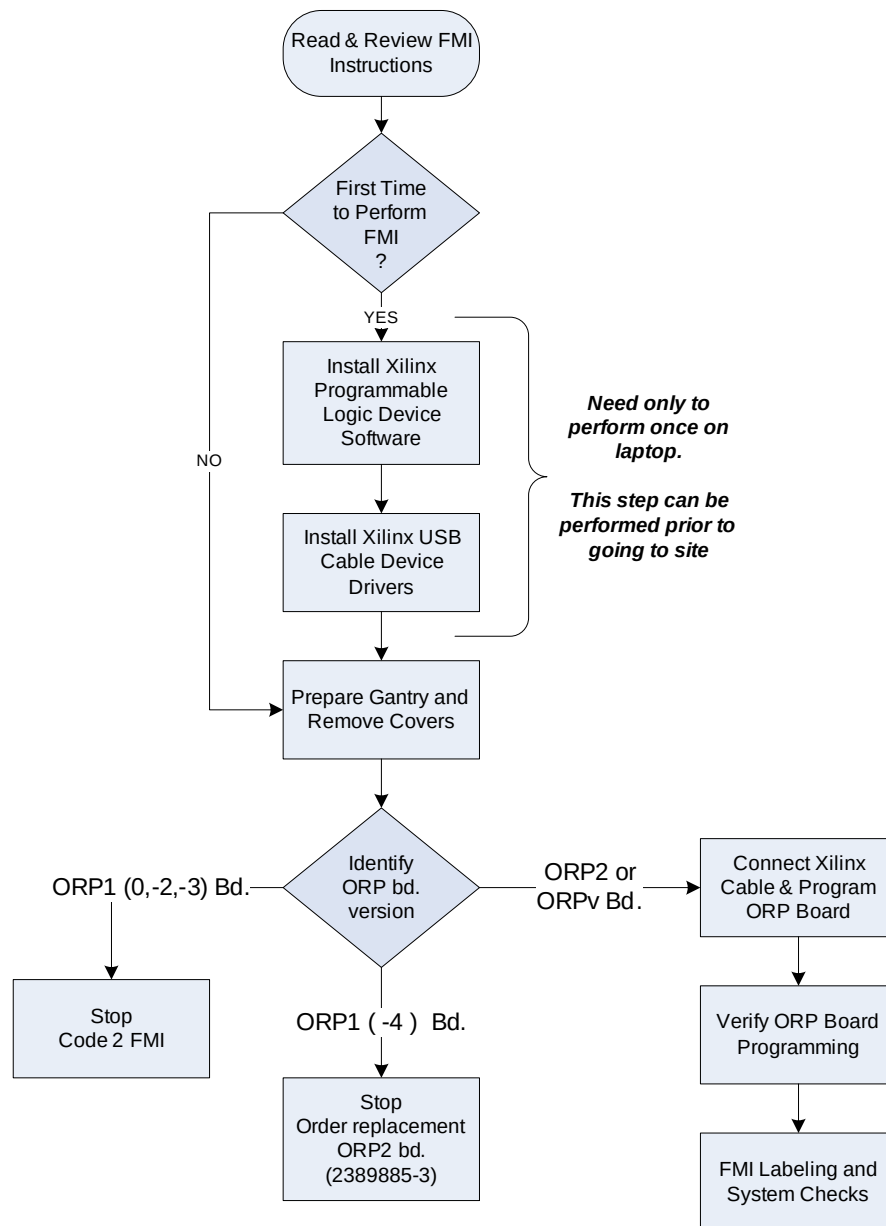
⚠ WARNING

POTENTIAL FOR SHOCK.

VOLTAGE MAY BE PRESENT.

USE PROPER LOCKOUT/TAGOUT PROCEDURES BEFORE REPLACING COMPONENTS.

FMI Procedure

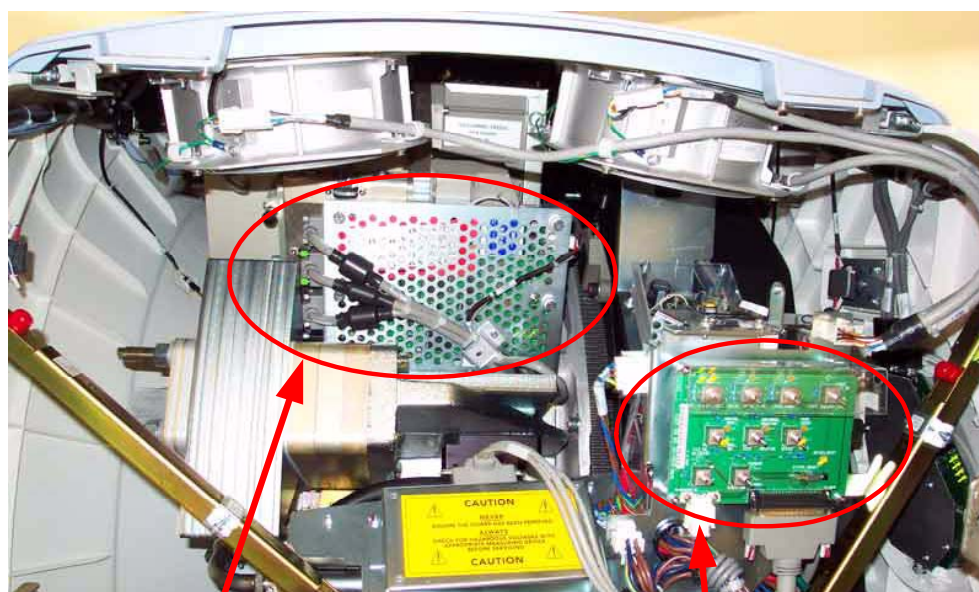


This flowchart is a high level view of this FMI procedure. Follow detail procedure below, but note where some sections do not need followed for each and every site. The Xilinx ISE Programmable Logic Device Software application needs installed only once on the FE's laptop.

I. Gantry Preparation and ORP Identification

Based on scan room layout and gantry access, decide which side of gantry would be easiest to gain access to ORP board. Preferably, access would be on the same side as the service switch panel.

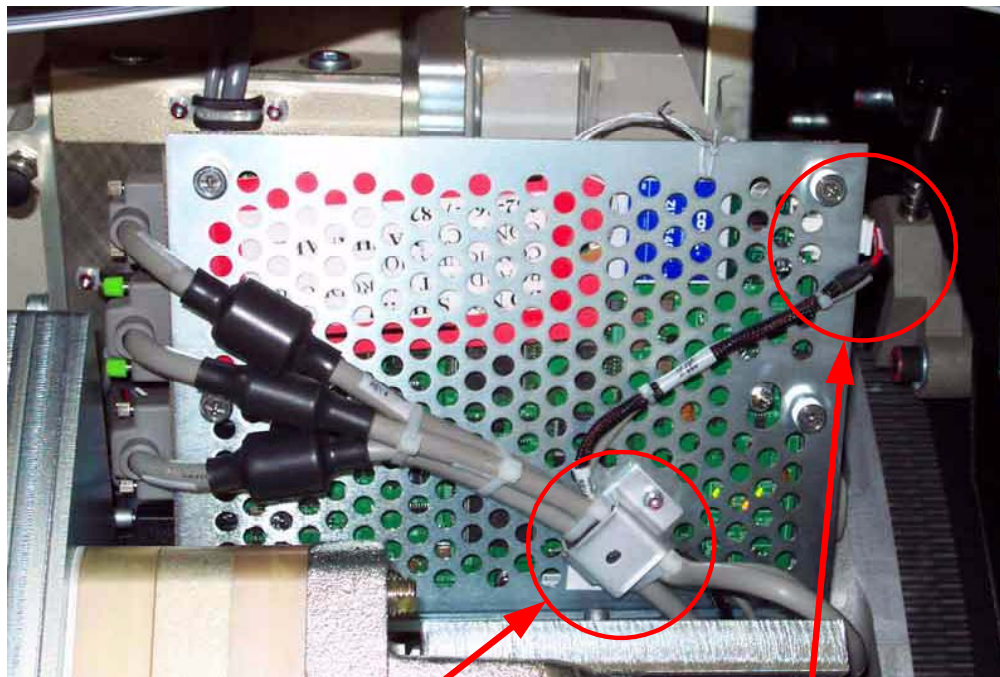
1. Remove gantry right side cover to gain access to service switch panel.



ORP Chassis

Service Switch Panel

2. Set gantry rotate angle so that ORP is accessible on the open side of gantry (~ 3 o'clock position if access is on right side).
3. Turn **off** Gantry 120V, Axial Drive and HVDC switches on the service switch panel. Use the Rotation Lock mechanism to keep rotate plate stationary.

**Cable Clamp****ORP Power Supply
Connector**

4. Remove cable clamp from ORP Chassis Cover and set aside. Take care not to drop socket cap screw and washers into gantry.
5. Slide up and remove ORP Chassis Cover, then set aside. You may find it necessary to disconnect the ORP power supply connector from chassis on upper right side.



6. Look at ORP board part number label. Check **part number** against list below.

Note: Do not reference the board name on the artwork of the Printed Circuit board or the "Rev #".

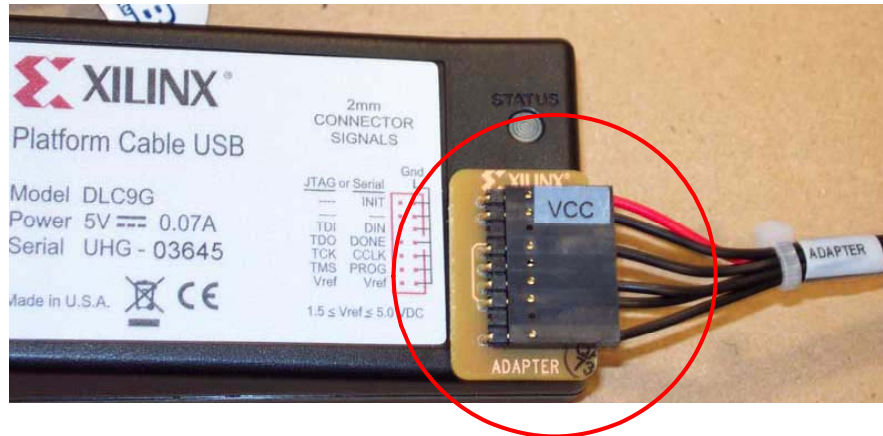
- | | | | |
|----|-----------|---|--------|
| a. | 2290254- | STOP, FMI Not required Close FMI Disp. as Code 2. | ORP1 |
| b. | 2290254-2 | STOP, FMI Not required Close FMI Disp. as Code 2. | ORP1-2 |
| c. | 2290254-3 | STOP, FMI Not required Close FMI Disp. As Code 2. | ORP1-3 |
| d. | 2290254-4 | STOP, Order replacement ORP2 (2389885-3)* | ORP1-4 |
| e. | 2389885- | FMI required Continue with Procedure! | ORP2 |
| f. | 2389885-2 | FMI required Continue with Procedure! | ORP2-2 |
| g. | 2389885-3 | STOP, FMI Not required, Fix has been done. Close FMI Disp. as Code 2. | ORP2-3 |
| h. | 5115622- | FMI required Continue with Procedure! | ORPV |
| i. | 5115622-2 | STOP, FMI Not required, Fix has been done. Close FMI Disp as Code 2. | ORPV-2 |

* Replacement of an ORP1-4 can be charged back to FMI 25392.

7. If system contains an "e, f, or h" board type, proceed onto ORP Flashing procedure in next section. (This is assuming the Xilinx ISE Programmable Device Logic Software has been loaded on laptop, otherwise see appendix I.)

II. ORP Flashing Procedure

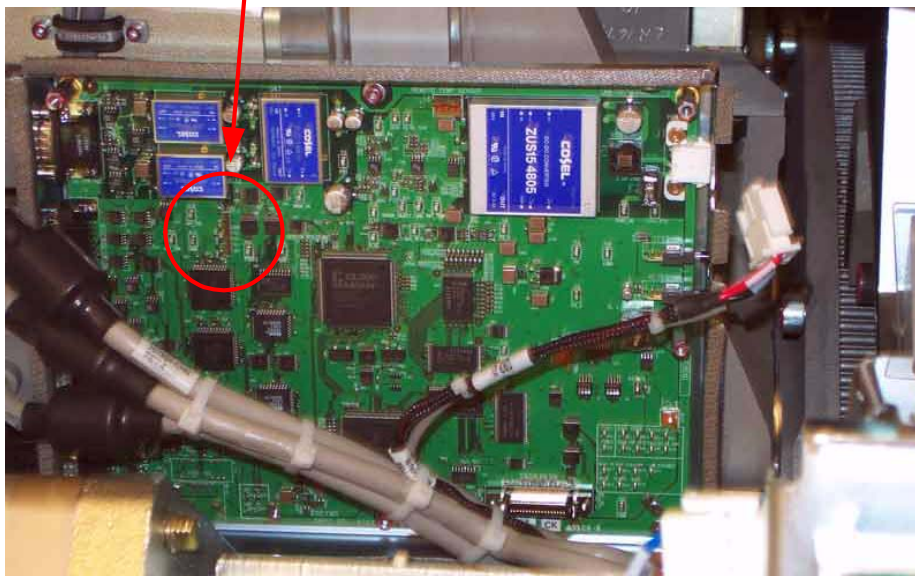
NOTE: If Xilinx ISE 8.2i Programmable Logic Device Software has not been previously loaded on your laptop, go to the Appendix (page 26) of this document and load it now! It was recommended in the Special Instructions on Cover Page to load prior to arrival on site!



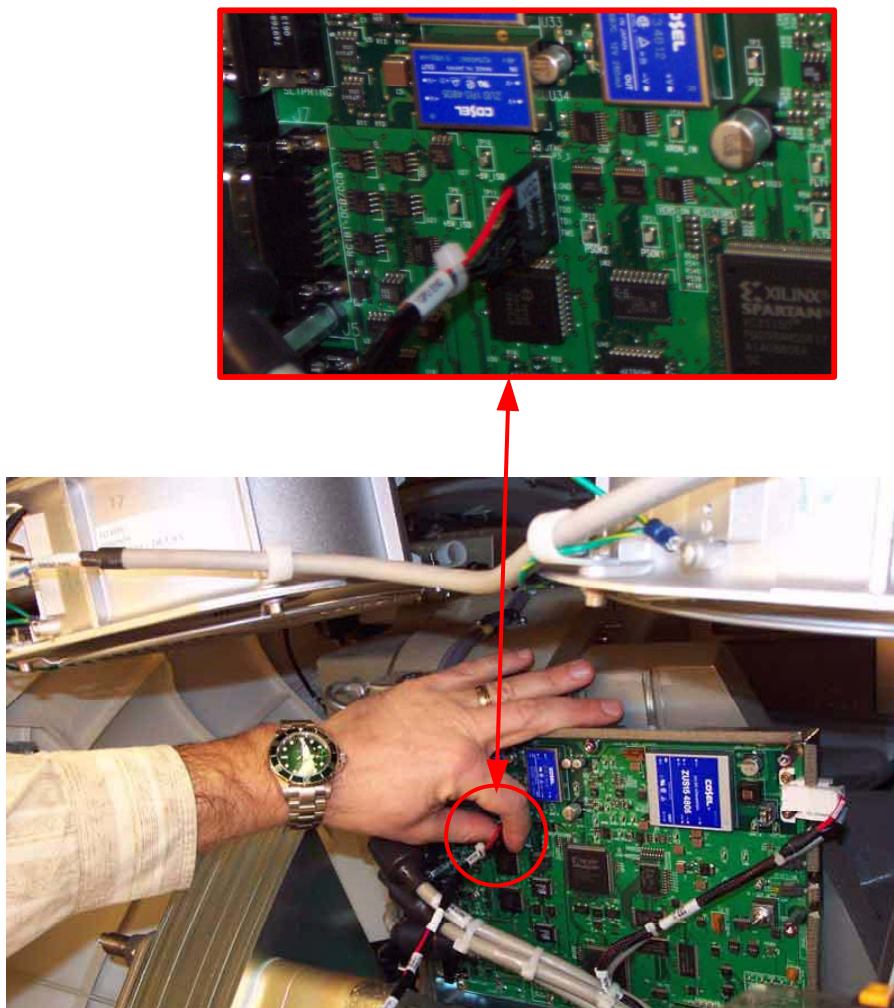
1. Connect the Interface Cable to the Xilinx Platform USB cable.

Caution: The VCC (Red) wire will be located at the top.

JTAG Header

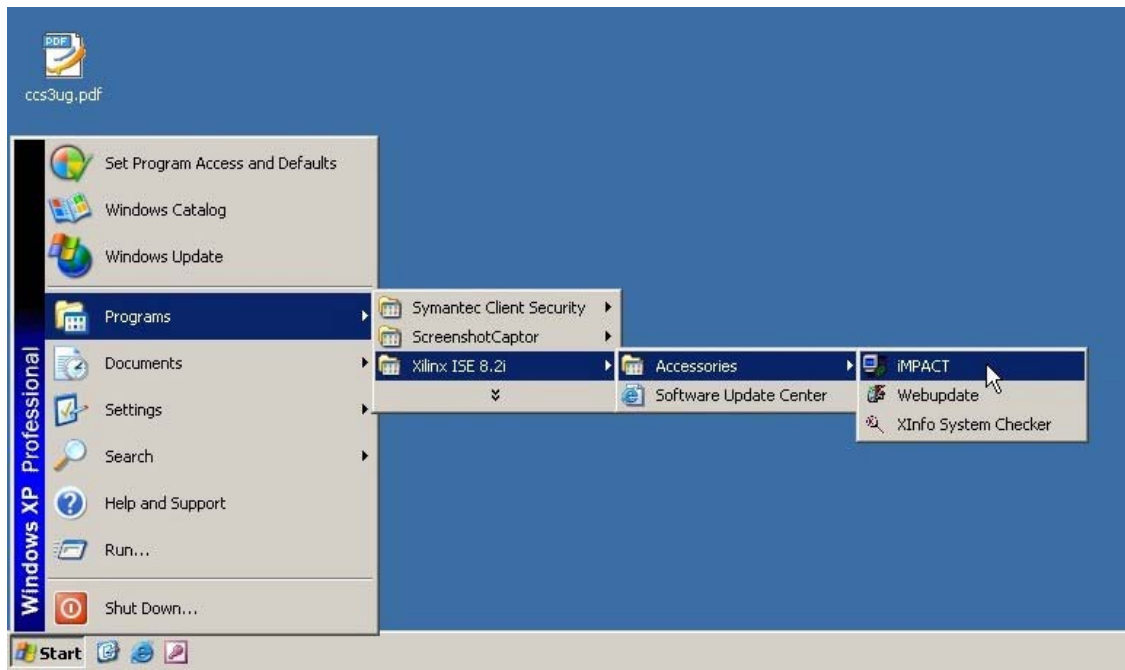


2. Connect the other end of the Interface Cable to the ORP board JTAG header.

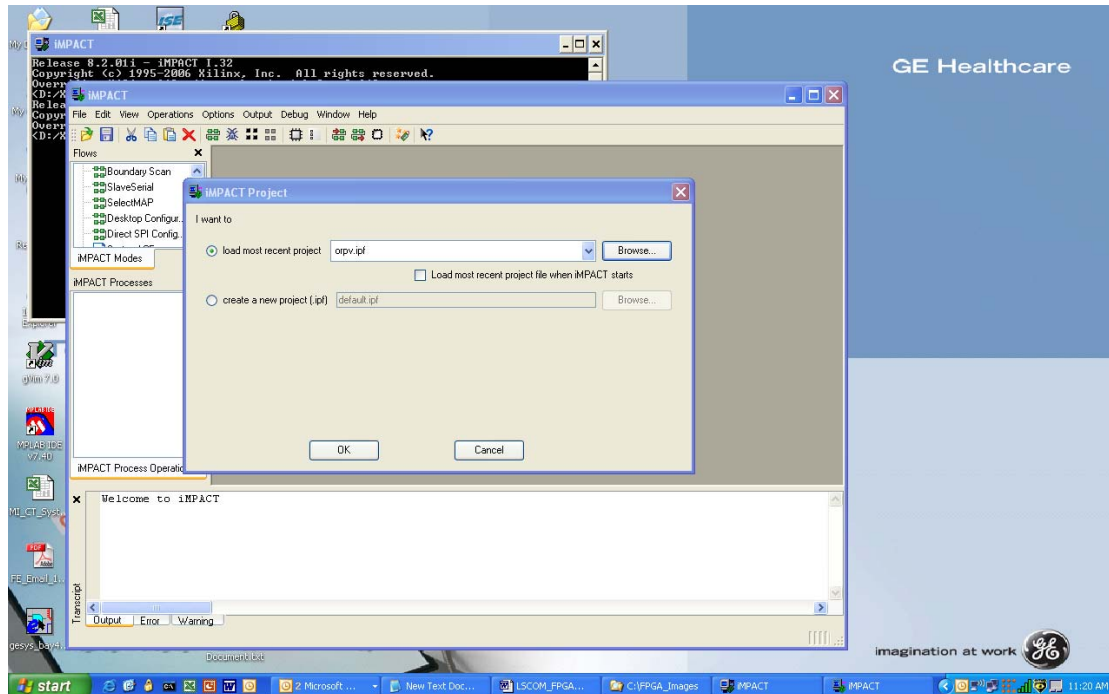


Caution: Pay attention to connector orientation. The VCC (Red) wire will be at the top.

3. Reconnect the ORP Power Supply connect if disconnected earlier.
4. Turn on the Gantry 120v on the Service Switch Panel



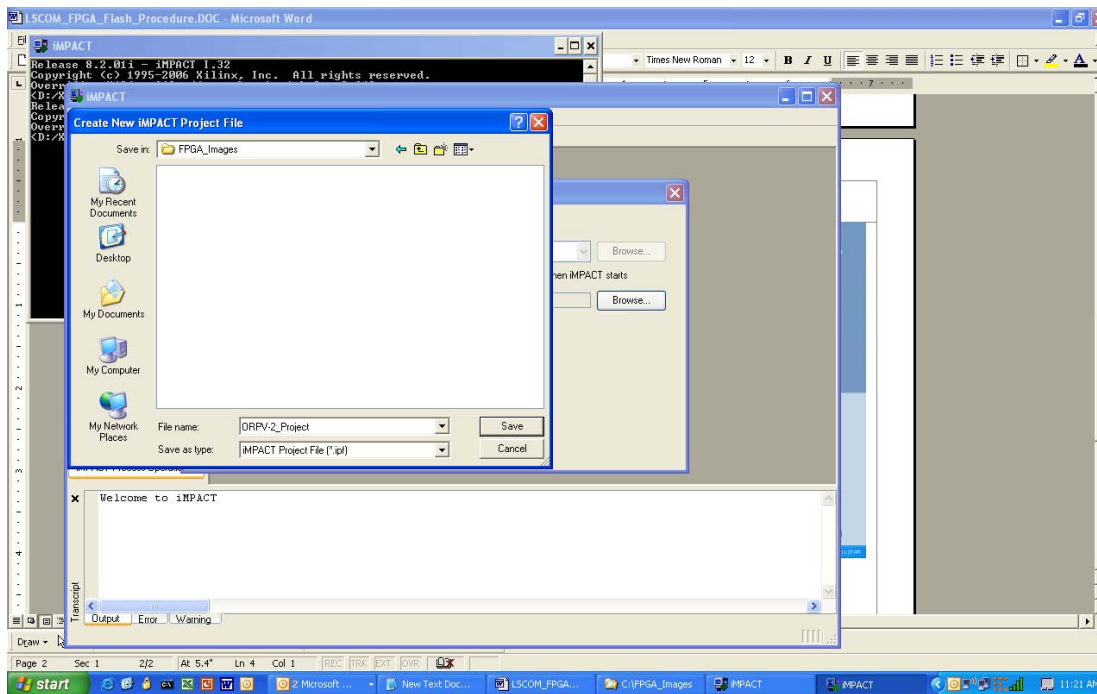
5. On the laptop, select the Xilinx Programmer Application program **iMPACT**.



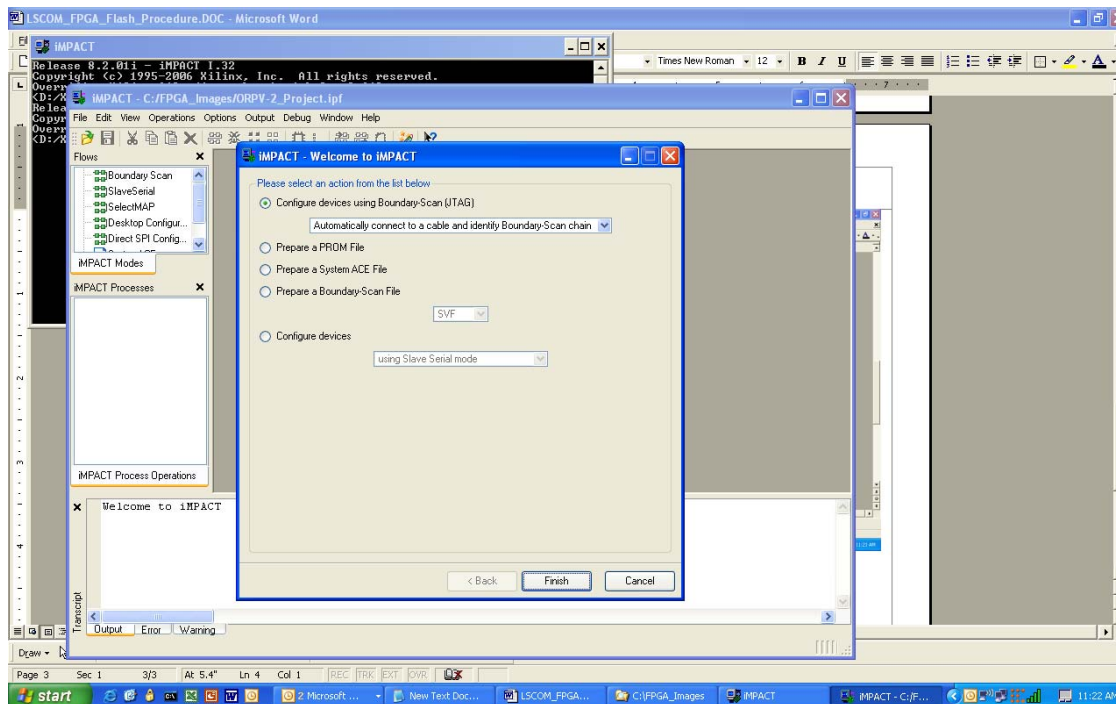
6. For first time execution of flashing procedure, select **create a new project (.ipf)**. Create a project name (i.e. ORPV or ORP2) and click **Browse**, and select a storage location.

Note:

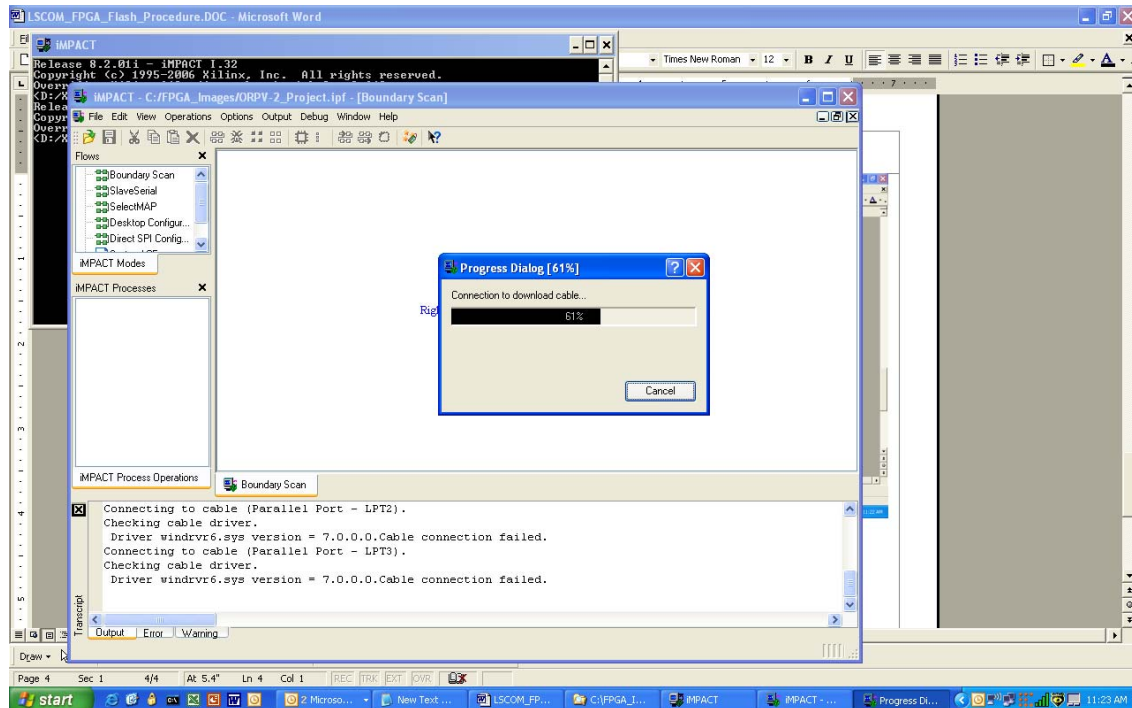
- *In future flashing of ORP boards due to this FMI, one may select the saved iMPACT Project files created in Step 6. This will save a few clicks. Just make sure to select the project file name that corresponds to the correct ORP board version.*
- *One must save separate project files for either an ORPV or an ORP2 board.*
- *If the flashing procedure has been executed on this laptop before, one may select **load most recent project**. Click Browse and select either ORPV or ORP2.ipf file that was saved from previous running of application. If using a previously saved project, **Skip to Step 14**.*



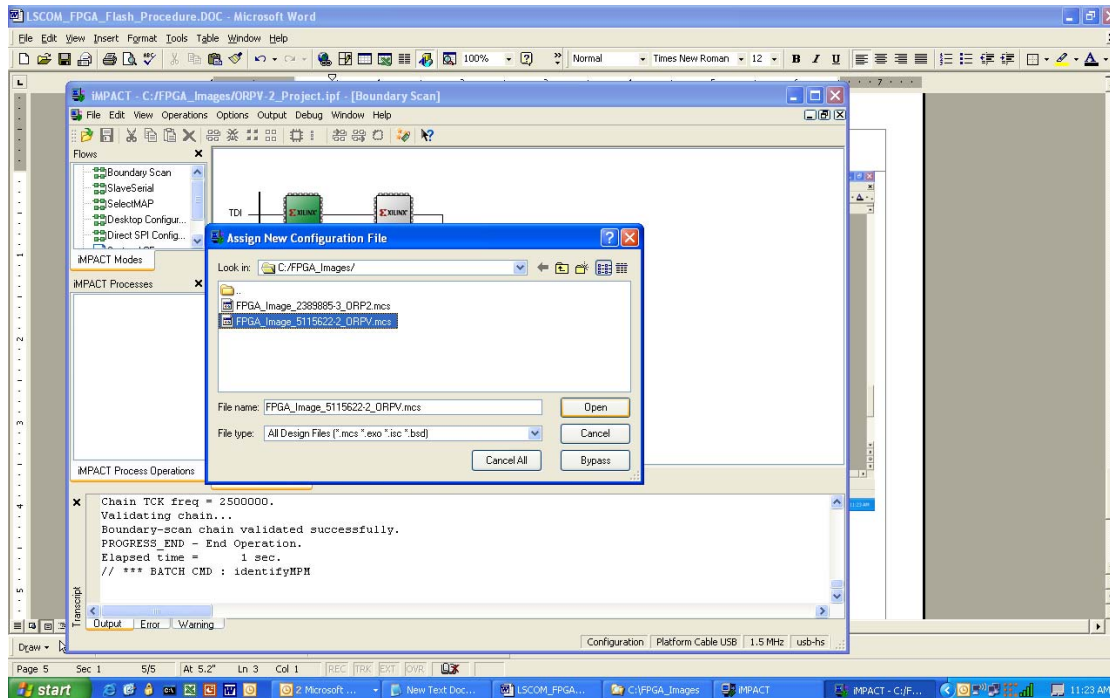
7. Click **Save**, and then **OK** on next window.



8. When this screen appears, keep default selections and click **Finish**.



9. The iMPACT program will now communicate with the ORP board.



10. The **Assign New Configuration File** pop-up window will appear. Change directory path to point to DVD/CD-ROM drive and select correct *.mcs file in pop-up window, click **OPEN**

Note: Make sure CD Rom is loaded in CD/DVD drive in laptop.

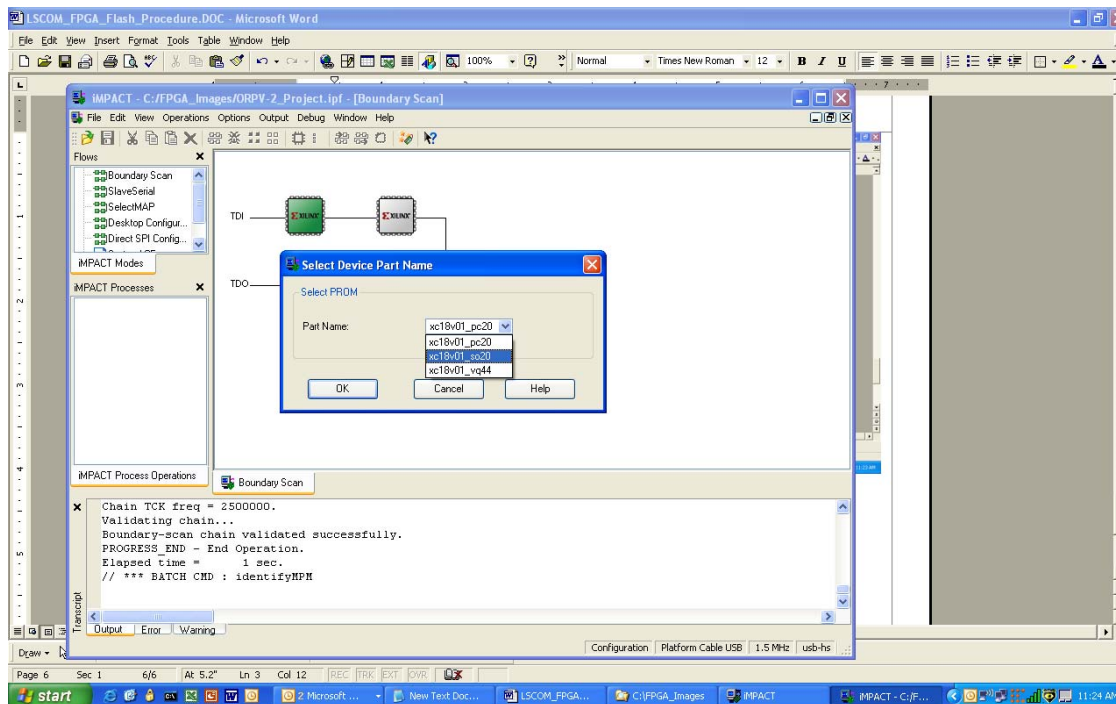
Select either the **ORPV** or **ORP2.mcs** file based on ORP identification conducted earlier.

File will be named:

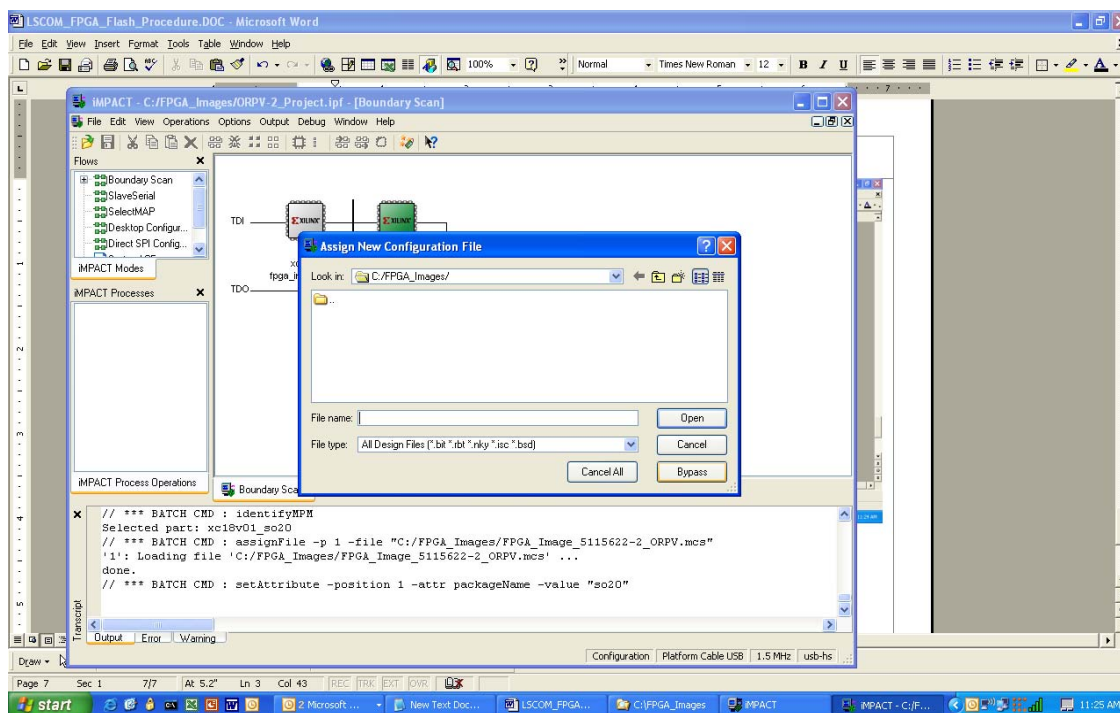
2389885-3_ORP2-3.mcs - for ORP 2 boards

Or

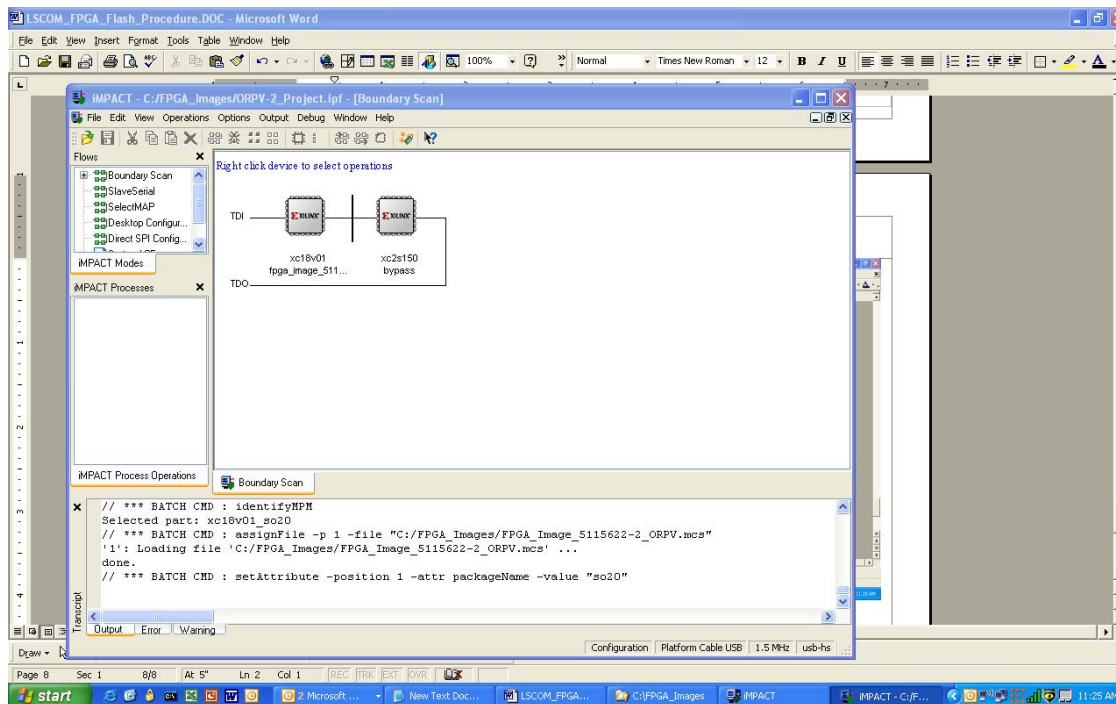
5115622-2_ORPV.mcs - for ORPV boards



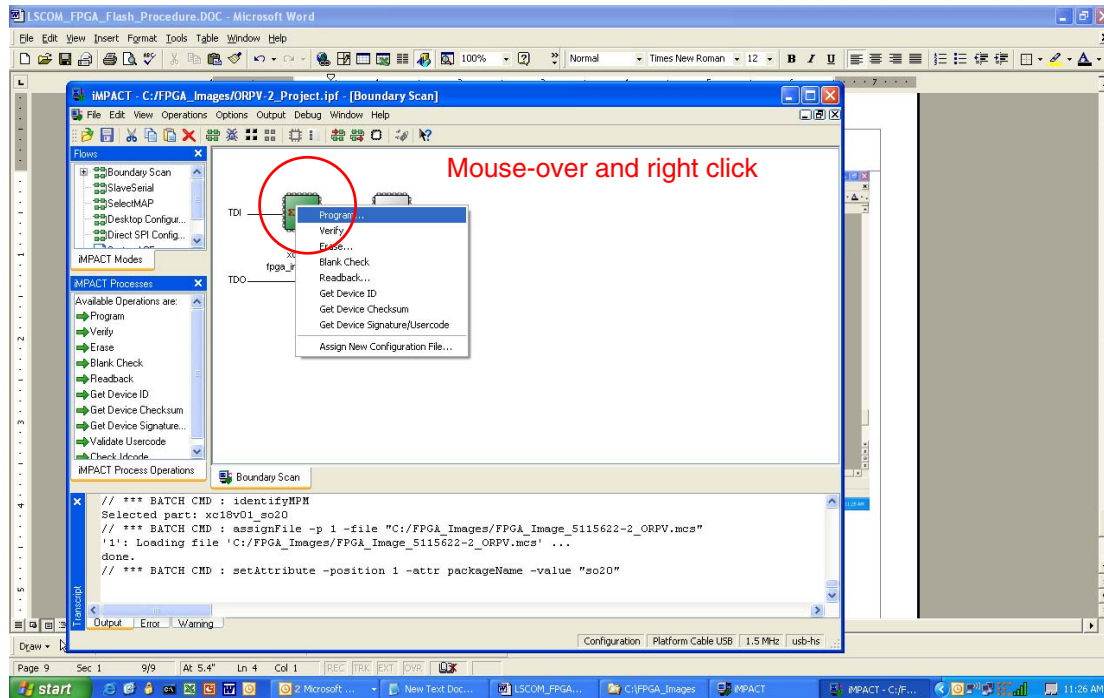
11. Click pull-down menu for **Part Name** on next pop-up window and select part ending with **so20**, and click **OK**.



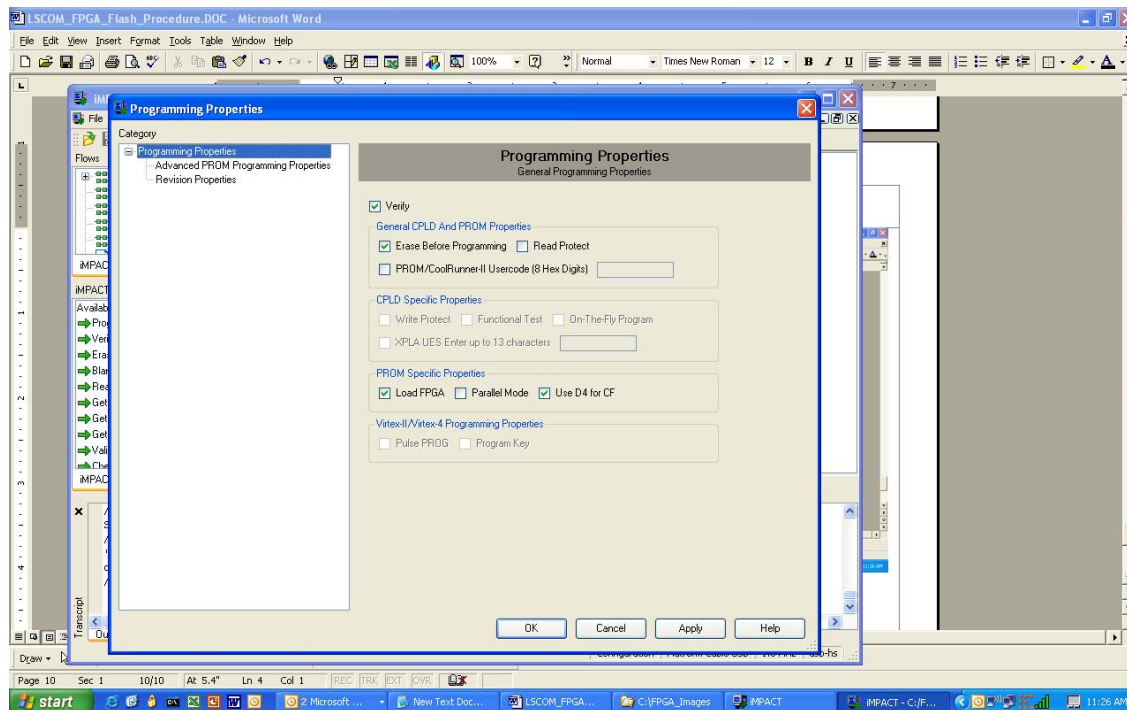
12. Click **Bypass** on next pop-up window.



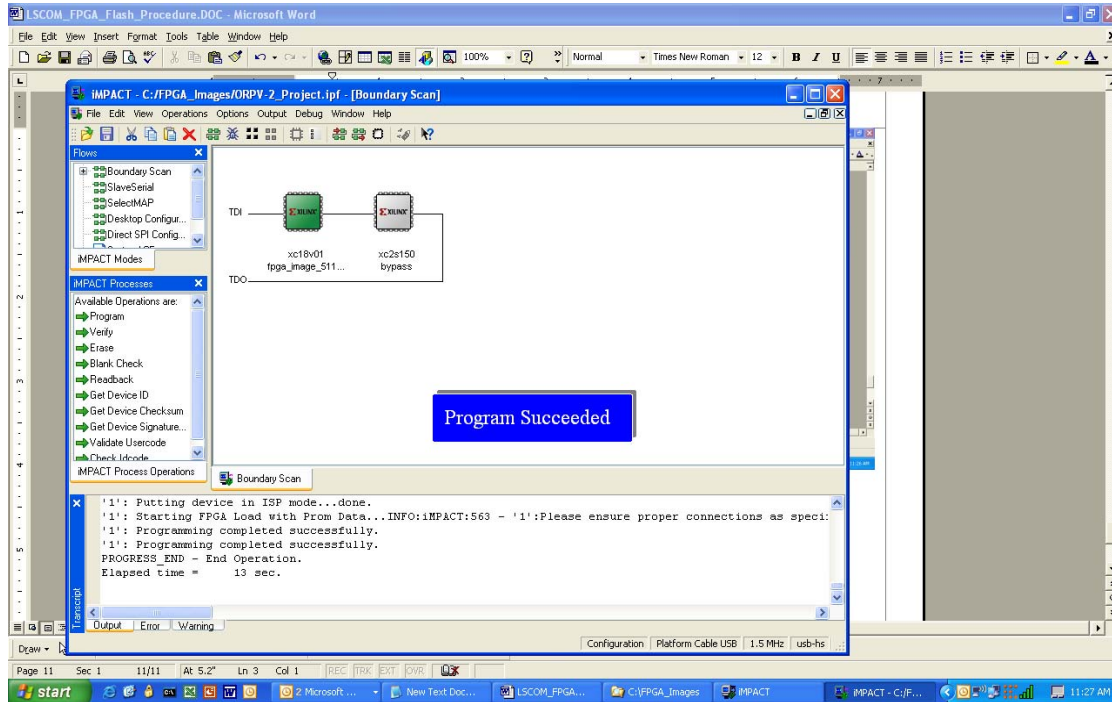
13. Screen will now look like the above.



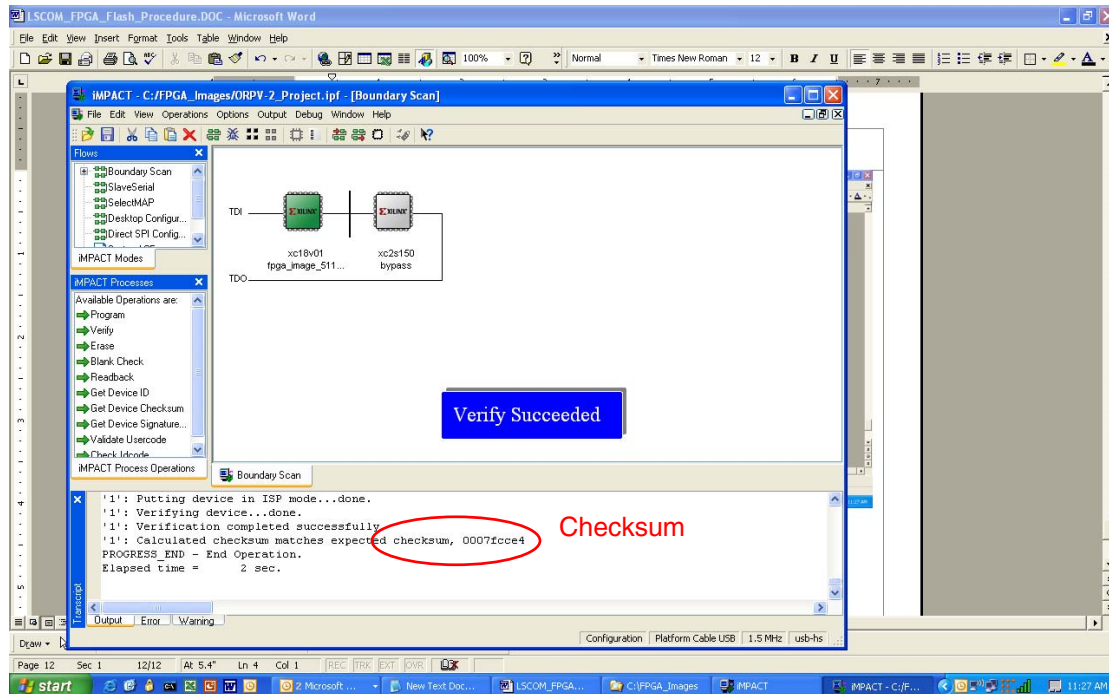
14. With mouse pointer on the first (left most) Xilinx device in window, right click and select **Program**.



15. Accept default selections in pop-up window and click **OK**.



16. The new flash code will be loaded onto the ORP board. Upon completion **Program Succeeded** will be displayed.



17. Right click on the first (left most) device again and select **Verify**. Upon completion **Verify Succeeded** will be displayed.
18. Verify the checksum values displayed on laptop match those expected.

ORPV (5115622-2) **0007fcce4**

Or

OPR2 (23899885-3) **0007fb154**

19. Flash programming of the ORP is now complete. Turn off gantry 120v and disconnect the Xilinx Platform USB and interface cable from ORP board.

20. Apply new Part Number Label to ORP board.

For ORPV Boards 5115622-2

For ORP2 Boards 2389885-3

Note: Take care to select the correct label and apply label over existing one on board.

21. Replace ORP Chassis Cover and Cable Clamp removed earlier. Check that all hardware is torqued to the correct value of 2.9N-m (25.66 in-lb).

Note: Take care not to hit fuse on the ORP board, located on the bottom left side, when fitting cover. The wires that connects to the ORP chassis should be tie wrapped together, then held in place with cable clamp. Tie wrapping the harness to the chassis cover is not necessary.

Finalization

1. Release Rotation Lock mechanism and turn **ON** Gantry 120V, Axial Drive and HVDC switches on the service switch panel.
2. Perform a Scan Reset to re-initialize ORP, CCB and TGP boards.
3. Replace gantry covers and perform test scan to confirm system performance before returning system for patient scanning.
4. Close FMI service dispatch per local procedures.

Note: As stated in the **Purpose Section** of this instruction:

"Due to frame timing code error in the Xilinx - Field Programmable Gate Array (LSCOM FPGA) of the ORP1-4, ORP2-2 and ORPV boards; a slip ring disconnect signal occurring along with a DAS trigger may result in an ORP command to turn off High Voltage at the Jedi generator."

Slip Ring Disconnect signals are also part of the problem. Though Slip Ring Disconnects are expected as part of normal operation, excessive disconnects are not (typically 1 disconnect per patient exam). Brush Block alignment and Slip Ring cleaning need to be checked if a site is suffering from kV Loss on a regular basis. Follow service procedures for aligning brush block and slip ring cleaning if required.

Leave the Xilinx ISE Programmable Logic Device Software and Programing Cable (FMI Kit) with system when finished. Secure with other system tools.

Appendix

I. Xilinx ISE 8.2i Programmable Logic Device Software Load Instructions

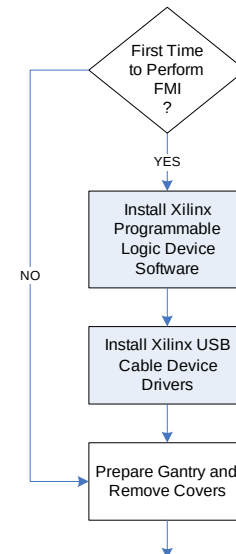
Workflow & Process Overview

Loading of the Xilinx ISE Programmable Logic Device Software application, required for flashing the ORP boards with the updated LSCOM code, needs only performed once on your laptop. For additional service dispatches for this FMI, these steps can be skipped.

The Xilinx Programmable Logic Device Software Application can be loaded on laptop at any time and it is advisable to load prior to arriving on site to minimize system down time.

This software requires a Windows 2000 or XP OS Laptop with USB port. (Supports both USB1 & 2)

The install process should take no more than 15 –20 minutes.



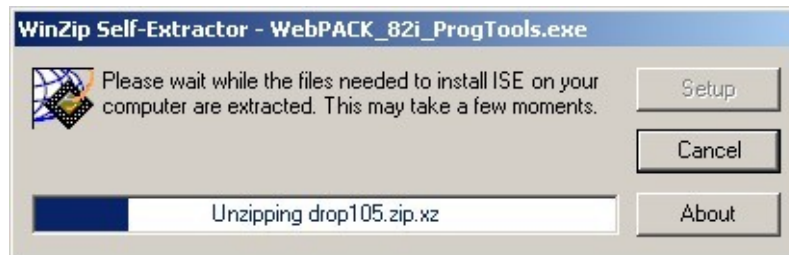
Caution: Do not connect the Xilinx Platform USB Cable supplied with FMI kit to the laptop prior to or during the application software load. The device drivers needed for this USB cable will be written to the laptop during this process.

Note: The following instructions are based on the software installation in the Windows XP environment. The wizard screens could be slightly different for Windows 2000 environments.

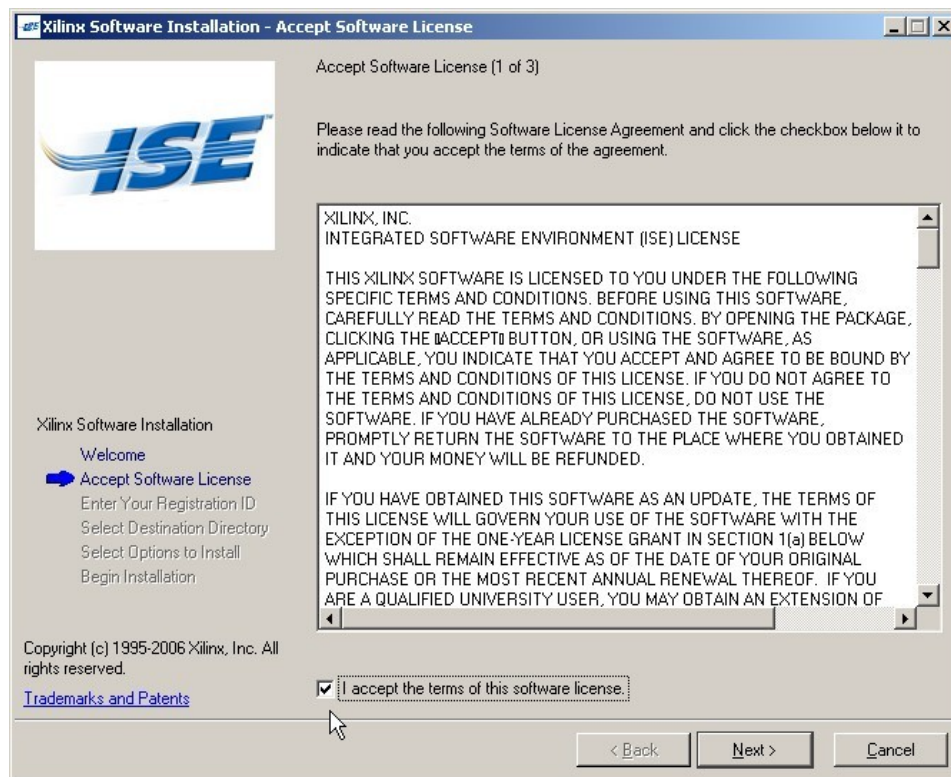
1. Place the CD-Rom disk supplied with the FMI kit in the laptop's DVD/CD-ROM drive.

Note: The CD does not have an "Autorun" feature

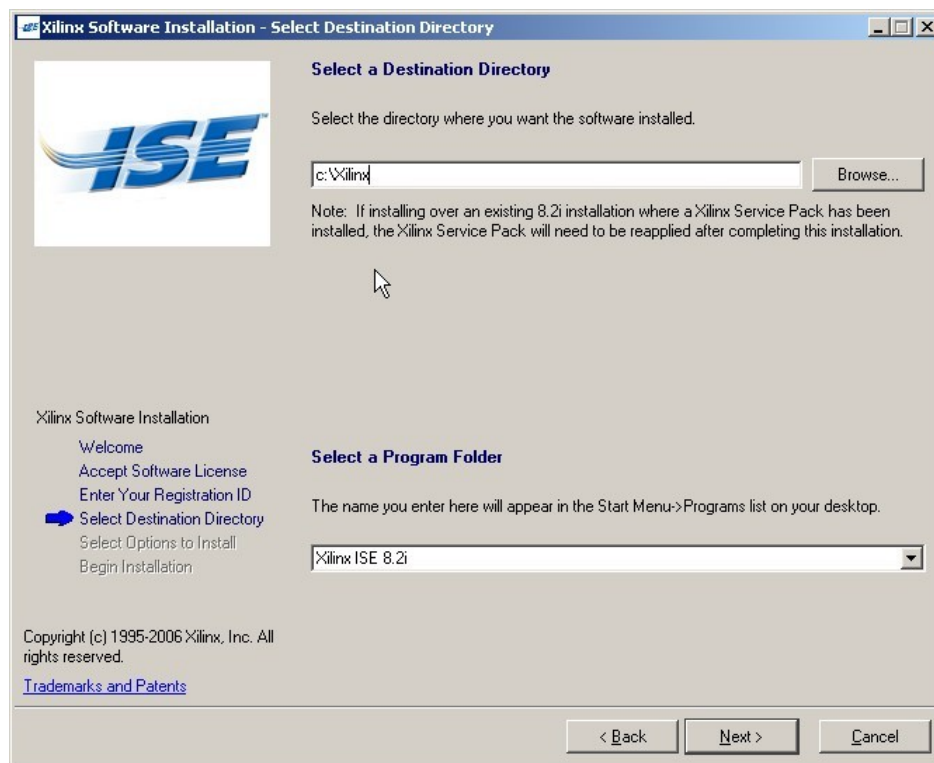
2. Open Windows Explorer or My Computer and navigate to the laptop's DVD/CD-ROM drive.
3. Select and execute **WebPACK_82i_ProgTools.exe** file on CD-ROM



4. Double click on file to execute program and begin installation.

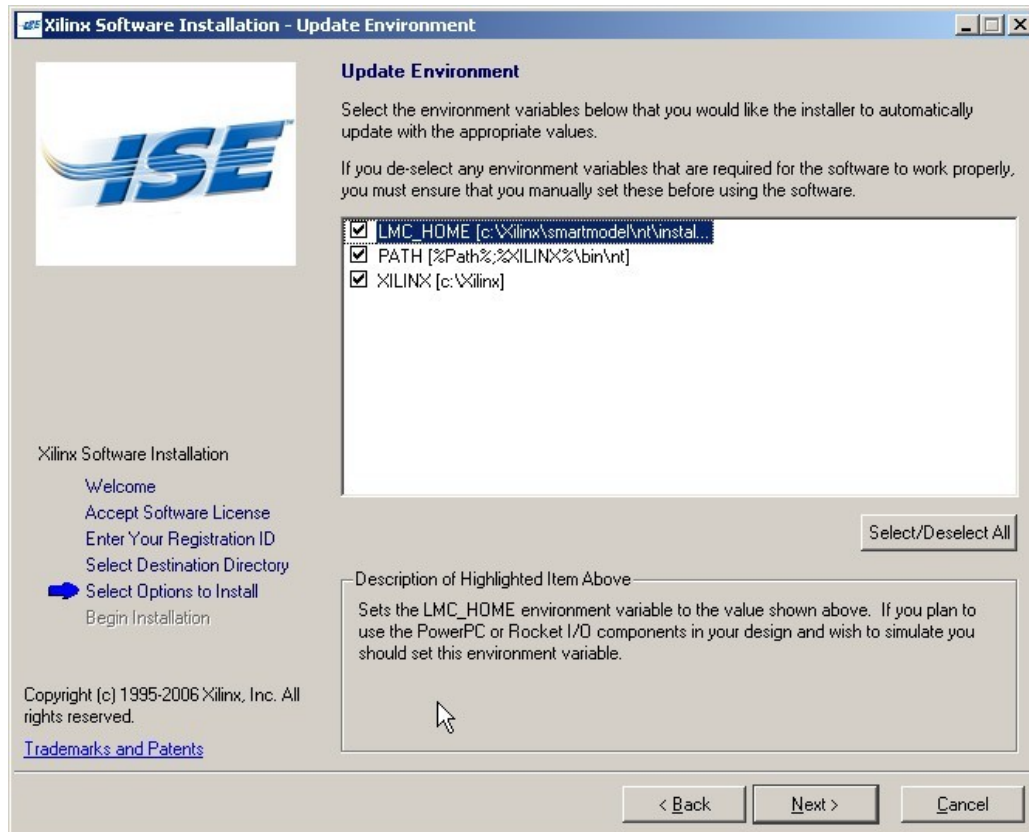


5. Accept the license agreement by selecting "I accept...." and click **Next**.
6. Repeat step 5 for next other two EULA's (End User License Agreements) that appear



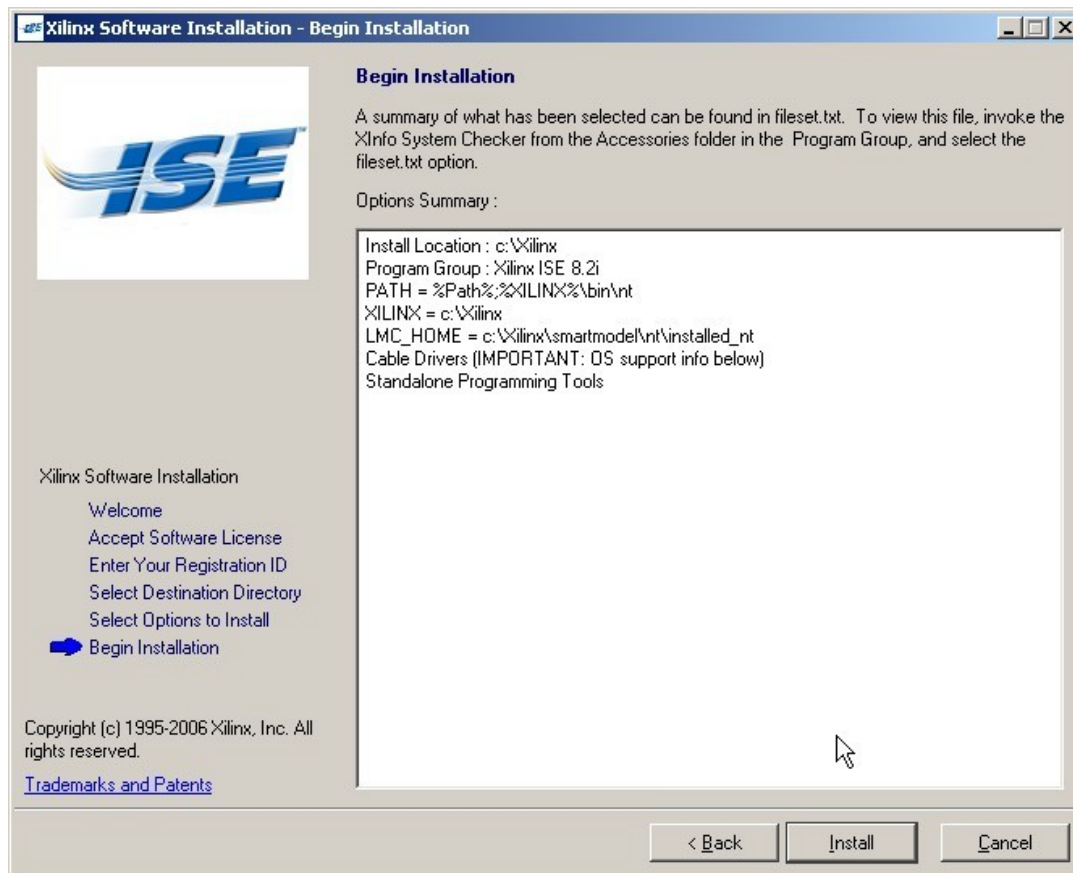
7. Accept the default entries in the next pop-up and click **Next**.

Note: One may change the default **Destination Directory** and **Program Folder** name if so desired. Please remember your changes!



- Accept default selection on next pop-up and click **Next**.

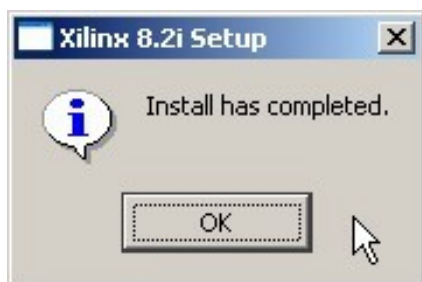
Note: Make no changes the default selections!



9. Click **Install** to begin software installation.



10. Xilinx's software will now load on the laptop; this should only take a few minutes.



11. Click **OK** when installation has finished.

12. **Reboot** laptop upon completion of application software load.

II. Xilinx USB Cable Device Driver Install

After successfully loading the Xilinx ISE Programmable Logic Device Software application software on the laptop, connect the Xilinx Platform USB Cable to any of the laptop's USB ports. The laptop should immediately see the new hardware and begin the device drive installation.

Note: Windows invokes the Found New Hardware flow each time Platform USB Cable is plugged into a different physical USB port for the first time. The wizard screens could be slightly different for Windows 2000 environments.



1. Connect the Xilinx USB cable to the Xilinx Platform box and to any USB port on laptop.



2. Select **Yes, this time only** and click **Next**.



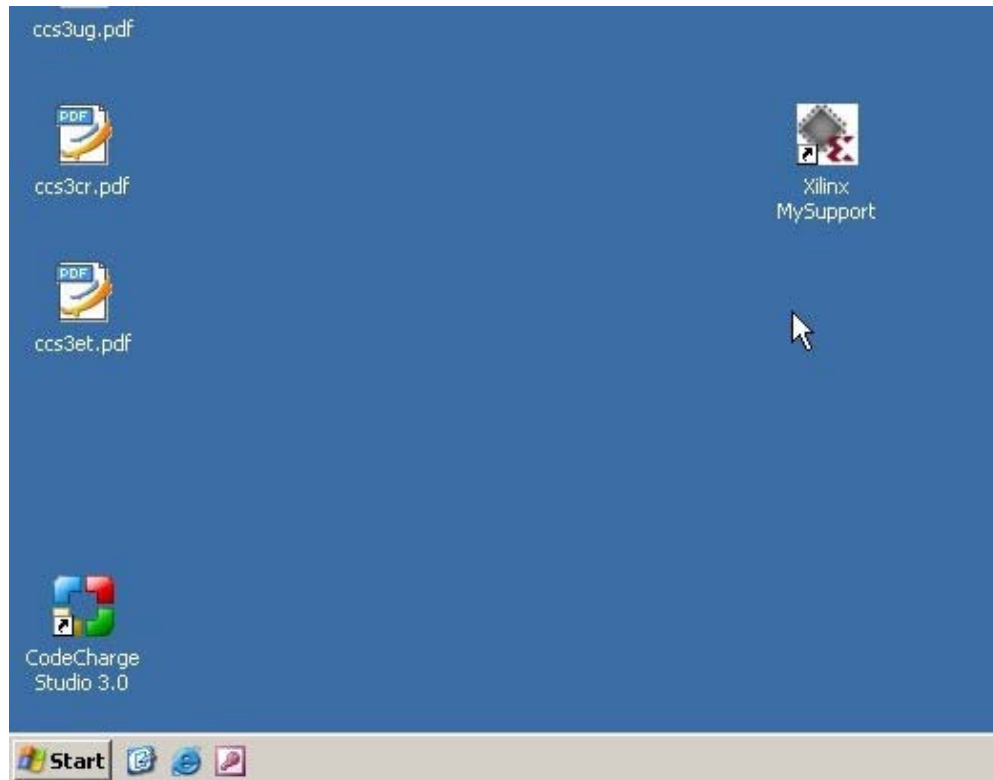
3. Select **Install the software automatically (Recommended)** and click **Next**.



4. Click **Finish** when the **Completing the Found New Hardware Wizard** pop-up appears.
5. Repeat steps 1-3, two more times. This will load the additional device drivers to support the Xilinx Platform USB cable.

Upon completion of the USB device drivers, the laptop and the associated Xilinx application and USB cable are now ready to perform the flash programming of the ORP boards.

III. Xilinx ISE 8.2i Programmable Logic Device Software Support



Note: The install process will place a new icon on the desktop. This icon will connect to Xilinx's support web site if Internet access is available.

If problems are encountered with loading the Xilinx ISE 8.2i Software, help can be found using the Search feature on the web site.

Though problems are not expected, occasionally missing or misplaced **Windows/System 32/** files may cause problems with execution of the Xilinx ISE 8.2i Software. Replace or move any file reported as missing.

Example: "iMPACT can not find msvcr71.dll in windows/system32/". The "msvcr71.dll" file is the C++ run time library file needed for this application software and is part of the standard software load of windows XP and 2000. On occasion some application software will delete or move System files from the Windows/System 32/ directory. Perform a global search of the laptop's hard drive for any missing files. Chances are these files are present in another directory. Simply copy the file and paste it into the Windows/System 32/ directory.